Rongqian Chen

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Education

George Washington University (GWU) — **Washington, D.C., US**
PhD in Electrical and Computer Engineering (GPA: 3.70) — *Sept. 2024 – Now*

University of Pennsylvania (UPenn) — **Philadelphia, US**
MSE in Electrical Engineering (GPA: 3.82) — *Sept. 2021 – May 2024*

Southwest Jiaotong University (SWJTU) — **Chengdu, China**
BEng in Automation (GPA: 3.45) — *Sept. 2017 – Jun. 2021*

Technical Skills

Robot Learning: Reinforcement Learning, Imitation Learning, Supervised fine-tuning, self-supervised Learning, VLA, WAM
Simulation: MuJoCo, Isaac Sim, Gazebo, ROS2
Hardware: Unity, KiCad, Altium Designer, SolidWorks, Ansys, PSIM, STM32, FPGA, DSP, Raspberry Pi, Arduino
Programming: Python,PyTorch, C, C++, C#, CMake, Bash, Git, Verilog, Matlab, SQL

Work Experience

DAXO Robotics: 120-DOF Dexterous Hand — **May 2026 – Now**
Robotics Research Intern – DAXO Robotics — *San Francisco, CA, US*

Hardware & Firmware
- Designed multiple revisions of **motor driver and communication PCBs** and **flexible PCB connectors** for a 120-DOF tendon-driven robotic hand in KiCad.
- Developed **STM32 firmware** for brushless motor drivers with **FOC control**, and built the full driver and control stack for real-time control of all 120 actuators.
- Performed **tendon-to-pose system identification** to map tendon actuation to finger joint poses.

Software & Robot Learning
- Built a **teleoperation data collection system**,**vision-based hand tracking** and **motion retargeting** for the 120-DOF hand to record synchronized demonstrations and maps human hand poses.
- Trained **imitation learning**& **reinforcement learning** policies in MuJoCo for **handwriting** with the tactile feedback high-DOF hand. Designed a **fine manipulation benchmark** in MuJoCo to evaluate performance.
- Fine-tuned and deployed the **GR00T N1.5** Vision-Language-Action (VLA) model on collected demonstrations.

Research Experience

Human-UAV Synergy System in Real-world Scenarios — **Sept. 2025 – Present**
Team Member – CHASE Project, GWU — *Washington, D.C., US*
- Designed and implemented a light weight VLA model on Jetson Nano Super GPU that deploy on the UAVs.
- Implemented a cooperative reasoning framework that infers human intent from observed behavior and turns high-level human instructions into mission goals and sequential plans based on perception data.
- Built **Isaac Lab and Gazebo** simulations to evaluate UAV localization and validate sim-to-real flight trajectories.

World Models and Planning for Computer-Use Agents — **Oct. 2025 – Present**
Core Researcher – GWU — *Washington, D.C., US*
- **JEPA-CUA:** Developing a **latent-space world model** for GUI space that rolls out multi-step action plans in latent space for **MPC-style** plan verification.
- **IntentScore:** Proposed an intent-aware action scorer trained on cross-OS trajectories; works as a plug-in harness with zero-shot cross-platform transfer.
- **Agent Alpha:** Built step-level **Monte Carlo Tree Search** with a custom alpha-UCT metric, comparison-driven evaluation and action chunking for long-horizon tasks.

ACDZero: Tree Search and RL for Automated Cyber Defense — **Jan. 2026 – Present**
Team Member – GWU — *Washington, D.C., US*
- Developed an autonomous defense agent that combines **Graph Neural Networks** with **Monte Carlo Tree Search** to learn policies in large, dynamic network environments.
- Used graph embeddings of network states for **value estimation and policy guidance**, enabling efficient search over massive action spaces.
- Outperformed RL baselines (**DQN, PPO**) in robustness and scalability on the **CAGE Challenge 4** benchmark.

Agentic AI for Autonomous Chip Physical Design **Dec. 2025 – Jan. 2026**
Core Contributor – GWU *Washington, D.C., US*

- Developed **OpenLayout**, an open-source multi-agent platform for VLSI physical design; five specialized agents (Planner, Worker, Debugger, Analyzer, Optimizer) produce DRC-clean, timing-closed layouts from gate-level netlists.
- Contributed to a **benchmark** for agents on commercial EDA tools at advanced nodes.

Tunable-Stiffness Actuators and Bio-inspired Robots **Nov. 2021 – July 2024**
Research Assistant – Sung Lab, GRASP, UPenn *Philadelphia, US*

- Designed and characterized a **pneumatic tunable-stiffness bellows actuator** for a monopedal robot.
- Used tunable leg stiffness to achieve **energy-efficient vertical hopping** across varying ground profiles.
- Design and control of a **tunable-stiffness coiled-spring actuator** for a flexible robot arm.

Publications

Robotics

- **Rongqian Chen**, Yiding Chen, Jijie Zhou, Tom Z. Jiahao (2026). Grip, Trace, Write: Force-Aware Imitation and Reinforcement Learning for High-DoF Robotic Handwriting. ***PhysicalWM at CoRL 2026***. [Website]
- **Rongqian Chen**, Jun Kwon, Kefan Wu, Wei-Hsi Chen (2025). Tunable Stiffness for Energy-Efficient Vertical Hopping in a Monopedal Robot Across Varying Ground Profiles. ***ICRA 2025***. [PDF]
- **Rongqian Chen**, Jun Kwon, Wei-Hsi Chen, Cynthia Sung (2024). Design and Characterization of a Pneumatic Tunable-Stiffness Bellows Actuator. ***RoboSoft 2024***. [PDF]
- Shivangi Misera, Mason Mitchell, **Rongqian Chen**, Cynthia Sung (2023). Design and Control of a Tunable-Stiffness Coiled-Spring Actuator. ***ICRA 2023***. [PDF]
- Zeyu Fang, Yuxin Lin, Cheng Liu, Beomyeol Yu, Zeyuan Yang, **Rongqian Chen**, Taeyoung Lee, Mahdi Imani, Tian Lan (2026). Uncertainty Mitigation and Intent Inference: A Dual-Mode Human-Machine Joint Planning System. ***HRD at IROS 2026***. [PDF]
- Zeyu Fang, Beomyeol Yu, Cheng Liu, Zeyuan Yang, **Rongqian Chen**, Yuxin Lin, Mahdi Imani, Tian Lan (2026). Reasoning Knowledge-Gap in Drone Planning via LLM-based Active Elicitation. ***AAAI 2026 Spring Symposium Series***.

Agents, Learning and Perception

- Xuan Wang, Jianan Jiang, **Rongqian Chen**, Tian Lan, Bin Li (2026). Too Hard, Too Easy, Just Right: Performance-Driven Challenge Adaptation in Virtual Reality Firefighter Training. [PDF]
- **Rongqian Chen**, Allison Andreyev, Xuan Wang, Jianan Jiang, Bin Li, Tian Lan (2026). From Description to Interactive VR Scene: An LLM Agent for Authoring Immersive Experiences. ***VIS × GenAI at IEEE VIS 2026***. [PDF]
- **Rongqian Chen**, Allison Andreyev, Yanming Xiu, Mahdi Imani, Bin Li, Maria Gorlatova, Gang Tan, Tian Lan (2025). Cognitive Attack Detection in Augmented Reality (CADAR): A Neuro-Symbolic Approach with Particle Filtering on Perception Graphs. ***CVPR 2026 Findings***. [PDF]
- Sizhe Tang, **Rongqian Chen**, Tian Lan (2026). Agent Alpha: Tree Search Unifying Generation, Exploration and Evaluation for Computer-Use Agents. ***COLM 2026***. [PDF]
- **Rongqian Chen**, Yu Li, Zeyu Fang, Sizhe Tang, Weidong Cao, Tian Lan (2026). IntentScore: Intent-Conditioned Action Evaluation for Computer-Use Agents. [PDF]
- Sizhe Tang, Guangyu Jiang, Yu Li, **Rongqian Chen**, Ioannis G Kevrekidis, Tian Lan (2026). FlowEdit: Information-Theoretic Control of LLM Reasoning Flows for Ill-posed Problems Involving Conflicts. ***MATH-AI at NeurIPS 2026***. [PDF]
- Li, Qiufeng, **Rongqian Chen**, Quan Cheng, Chengxuan Wang, Sizhe Tang, Wuxi Li, Duo Ding et al (2026). PDAGENT-BENCH: Characterizing, Grounding, and Architecting LLM Agents for VLSI Physical Design. [PDF]
- Yu Li, Sizhe Tang, **Rongqian Chen**, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D Bastian, Tian Lan (2026). ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense. ***ICCN at INFOCOM 2026***. [PDF]
- Shu Hong, **Rongqian Chen**, Rifatul Islam, Mahdi Imani, Gang Tan, Tian Lan (2025). Time-Aware LSTM for Gaze Prediction in Mixed Reality Under Latency Perturbations. ***XR Security at ACM MobiHoc 2025*** (Best Poster Award). [PDF]
- **Rongqian Chen**, Shu Hong, Rifatul Islam, Mahdi Imani, Gang Tan, Tian Lan (2025). Perception Graph for Cognitive Attack Reasoning in Augmented Reality. ***XR Security at ACM MobiHoc 2025***. [PDF]

Earlier Work on Electrical Systems

- **Rongqian Chen**, Yingquan Zou, Anyong Gao, Leshi Chen (2022). A Cluster-Based Weighted Feature Similarity Moving Target Tracking Algorithm for Automotive FMCW Radar. ***VTC 2022-Spring***. [PDF]
- Ping Yang, Xi Chen, **Rongqian Chen**, et al (2021). Stability Improvement of Pulse Power Supply with Dual-Inductance Active Storage Unit Using Hysteresis Current Control. ***IEEE JETCAS 2021***. [PDF]